

3. Constructors

- ❖ Constructor are the special method whose name is same a class name.
- ❖ Constructor does not have return type even void also.
- ❖ Constructor will be invoked by the JVM automatically at the time of object creation.
- ❖ 2Constructors are mainly used to initialize instance variables of class with different set of values.
- ❖ The Constructor without any arguments is called as Default Constructor.
- ❖ When you are not writing any constructor inside the class then one default constructor will be inserted by Java compiler.
- ❖ When you are writing any constructor inside the class by changing parameter. This process is called as CONSTRUCTOR OVERLOADING.
- ❖ When you are overloading Constructor then parameters must differ in terms of
 - ⇒ Number of parameter
 - ⇒ Type of parameter
 - ⇒ Order of parameter

Example:

```
class Hello {
    int a;
    String str;
    Hello (int a, String str) {...} //VALID
    Hello (int a, String str) {...} //INVALID
    Hello (int a, int b) {...} //VALID
    Hello (String str, int a) {...} //VALID
}
```

- ❖ You cannot invoke the constructor Always JVM invokes automatically.
- ❖ You cannot specify the return type for the constructor .When you specify the return type for the constructor then it will be treated as normal method.
- ❖ You can use **empty return** statement inside the constructor.
- ❖ The process of invoking one constructor from another constructor using this is called as CONSTRUCTOR CHAINNG.

Programs:

OOPS56.java

```
class OOPS56{
    public static void main(String args[]){
        Student stu1=new Student();
        stu1.show();
        Student stu2=new Student();
        stu2.show();
    }
}
class Student{
    int sid;
    String sname;
    void show(){
        System.out.println(sid+" \t "+sname);
    }
}
```

OOPS57.java

```
class OOPS57{
    public static void main(String args[]){
        Student stu=new Student();
        stu.sid=99;
        stu.sname="kumar";
        stu.show();
    }
}
class Student{
    int sid;
    String sname;
    void show(){
        System.out.println(sid+" \t "+sname);
    }
}
```

OOPS58.java

```

class OOPS58{
    public static void main(String args[]){
        Student stu=new Student();
        stu.sid=99;
        stu.sname="Kumar";
        stu.show();
    }
}
class Student{
    int sid;
    String sname;
    Student() {
        System.out.println("Student def cons");
    }
    void show(){
        System.out.println(sid+" \t "+sname);
    }
}

```

OOPS59.java

```

class OOPS59{
    public static void main(String args[]){
        Student stu=new Student();
        stu.sid=99;
        stu.sname="Kumar";
        stu.show();
    }
}
class Student{
    int sid;
    String sname;
    Student(int id, String sn){
        System.out.println("Student 2-Arg Cons");
        sid=id;
        sname=sn;
    }
    void show(){
        System.out.println(sid+" \t "+sname);
    }
}

```

OOPS60.java

```

class OOPS60{
    public static void main(String args[]){
        Student stu1=new Student(88,"Sai");
        stu1.show();
        Student stu2=new Student(99,"Ram");
        stu2.show();
    }
}
class Student{
    int sid;
    String sname;
    Student(int id, String sn) {
        System.out.println("Student 2-Arg Cons");
        sid=id;
        sname=sn;
    }
    void show() {
        System.out.println(sid+" \t "+sname);
    }
}

```

OOPS61.java

```

class OOPS61{
    public static void main(String args[]){
        Student stu1=new Student();
        stu1.show();
        Student stu2=new Student(99,"Ram");
        stu2.show();
    }
}
class Student{
    int sid;
    String sname;
    Student() {
        System.out.println("Student Default Cons");
    }
    Student(int id, String sn) {
        System.out.println("Student 2-Arg Cons");
        sid=id;
        sname=sn;
    }
    void show(){
        System.out.println(sid+" \t "+sname);
    }
}

```

OOPS62.javaOOPS63.java

```

class OOPS62{
    public static void main(String args[]){
        Student stu1=new
        Student(99,"Om","om@",999);
        stu1.show();
        Student stu2=new Student(88,"Sai","sai@");
        stu2.show();
        Student stu3=new Student(77,"Ram");
        stu3.show();
        Student stu4=new Student();
        stu4.show();
    }
}
class Student {
    int sid;
    String sname;
    String email;
    long phone;
    Student(int id,String sn,String em,long ph) {
        System.out.println("Student 4 arg Cons");
        sid=id;
        sname=sn;
        email=em;
        phone=ph;
    }
    Student(int id, String sn, String em){
        System.out.println("Student 3 arg Cons");
        sid=id;
        sname=sn;
        email=em;
    }
    Student(int id, String sn)    {
        System.out.println("Student 2 arg Cons");
        sid=id;
        sname=sn;
    }
    Student()    {
        System.out.println("Default Cons");
    }
    void show() {
        System.out.println(sid+" \t "+sname);
    }
}

```

```

class OOPS63{
    public static void main(String args[]){
        Student stu3=new Student(77,"Om");
        stu3.show();
    }
}
class Student{
    int sid;
    String sname;
    Student(int id,String sn){
        System.out.println("Student 2 arg
        Cons");
        sid=id;
        sname=sn;
    }
    Student(int x,String y){
        System.out.println("Student 2 arg Cons");
        sid=x;
        sname=y;
    }
    void show(){
        System.out.println(sid+" \t "+sname);
    }
}

```

OOPS64.java

```

class OOPS64{
    public static void main(String args[]){
        Student stu3=new Student(77,"Om");
        stu3.show();
    }
}
class Student{
    int sid;
    String sname;
    Student(int id, String sn){
        System.out.println("Student 2 arg Cons");
        sid=id;
        sname=sn;
    }
    Student(String y, int x){
        System.out.println("Student 2 arg Cons");
        sid=x;
        sname=y;
    }
    void show(){
        System.out.println(sid+" \t "+sname);
    }
}

```

OOPS65.java**OOPS66.java**

```

class OOPS65{
public static void main(String args[]){

Student stu=new Student();
stu.Student(77,"Om");
stu.show();
}
}

class Student {
    int sid;
    String sname;
    Student(int id, String sn){
        System.out.println("Student 2 arg Cons");
        sid=id;
        sname=sn;
    }

    Student() {
        System.out.println("Student Default Cons");
    }
    void show() {
        System.out.println(sid+ " \t "+sname);
    }
}

```

```

class OOPS66{
public static void main(String args[]){

Student stu=new Student();
stu.Student(77,"SD");
stu.show();
}
}

class Student {
    int sid;
    String sname;
    void Student(int id, String sn){
        System.out.println("Student 2 arg Cons");
        sid=id;
        sname=sn;
    }

    Student() {
        System.out.println("Student Default Cons");
    }
    void show(){
        System.out.println(sid+ " \t "+sname);
    }
}

```

OOPS67.java

```

class OOPS67{
public static void main(String args[]){
Student stu=new Student (77,"Om");
stu.show();
}
}

class Student {
    int sid;
    String sname;
    void Student(int id, String sn) {
        System.out.println("Student 2 arg Cons");
        sid=id;
        sname=sn;
    }
    void show() {
        System.out.println(sid+ " \t "+sname);
    }
}

```

OOPS68.java

```

class OOPS68{
public static void main(String args[]){
Student stu=new Student(-12);
stu.show();
}
}

class Student {
    int age=18;
    Student (int ag){
        System.out.println("Student 1 arg Cons");
        age=ag;
    }
    void show() {
        System.out.println(age);
    }
}

```

OOPS69.java**OOPS70.java**

```

class OOPS69{
    public static void main(String args[]){
        Student stu=new Student(-12);
        stu.show();
    }
}

class Student{
    int age=18;
    Student (int ag){
        System.out.println("Student 1 arg Cons");
        if (ag<18)
            return;
        age=ag;
    }
    void show() {
        System.out.println(age);
    }
}

```

```

class OOPS70{
    public static void main(String args[]){
        Student stu=new Student(-12);
        stu.show();
    }
}

class Student{
    int age=18;
    Student (int ag){
        System.out.println("Student 1 arg Cons");
        if (ag<18)
            return 0;
        age=ag;
    }
    void show() {
        System.out.println(age);
    }
}

```

Program output /error and details:-

Prog No.	Output	Error and Details
OOPS56	0 null 0 null	Constructor does not have return type even void also. When you are not writing any constructor inside the class then one default constructor will be inserted by Java compiler. So here, class loaded then object created then method call.
OOPS57	99 Kumar	Here, updating value with class reference then call the show method.
OOPS58	Student def cons 99 Kumar	Here student () is Default Constructor because it does not have any arguments.
OOPS59	Compilation error	[Error: The constructor Student () is undefined]. Here, Student 2-Argument constructor likes int and String but we are not passing any argument with object creation.
OOPS60	Student 2-Arg Cons 88 Sai Student 2-Arg Cons 99 Ram	Here, Student 2-Argument constructor likes int and String but we are passing two arguments with object creation.
OOPS61	Student Default Cons 0 null Student 2-Arg Cons 99 Ram	Here, 1 st Student () is Default Constructor and 2 nd Students 2-Argument constructor. 1 st time we are not passing any argument with object creation but 2 nd time passing two arguments with object creation. it is called as Constructor overloading.

Prog No.	Output	Error and Details
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OOPS62	Student 4 arg Cons 1 Om Student 3 arg Cons 2 Sai Student 2 arg Cons 3 Ram Default Cons 0 null	When you are overloading Constructor then number of parameters must differ and you are printing only sid and sname so that it printing only two values.
OOPS63	Compilation error	[Error: Constructor Student (int, String) is already defined in class Student], overloading Constructor parameters must differ
OOPS64	Student 2 arg Cons 77 Om	When you are overloading Constructor then order of parameters must different. 2 nd Constructor will not execute due to value not matching.
OOPS65	Compilation error	[Error: cannot find symbol] because, we cannot initialize value of constructor with class reference.
OOPS66	Student Default Cons Student 2 arg Cons 77 Om	We cannot initialize value of method with class reference.
OOPS67	Compilation error	[Error: The constructor Student () is undefined].you can't use any return type for constructor.
OOPS68	Student 1 arg Cons -12	Constructors are mainly used to initialize instance variables of class with different set of values.
OOPS69	Student 1 arg Cons 18	You can use empty return statement inside the constructor.
OOPS70	Compilation error	[Error: cannot return a value from method whose result type is void or return o]

Write your note here.....